

A Family of Mathematical Constants Indexed by Gaussian Primes

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March 2026

Abstract

We identify an infinite family of mathematical constants, one for each Gaussian prime π of the ring $\mathbb{Z}[i]$ of Gaussian integers, each verified absent from the OEIS, Wolfram Alpha, and ISC databases at the precision tested. The constant associated to π is $c_\pi = |\text{Im}(z)/\text{Re}(z)|$ where $z = \Gamma(1+w/2)/\Gamma(1/2+w/2)$ and $w = \pi/N(\pi)$. Constants arise from split primes $p \equiv 1 \pmod{4}$ and from the ramified prime 2, but not from inert primes $p \equiv 3 \pmod{4}$. Conjugate primes $\pi, \bar{\pi}$ give equal $|c_\pi|$ but opposite angles $S(\pi) = -S(\bar{\pi})$. Among the Gaussian units, $S_i + S_{-i} = \pi/4$ is proved via hyperbolic identities. Among associates of the ramified prime above 2, the associates $\pi \in \{-1+i, -1-i\}$ satisfy $S_\pi + S_{\bar{\pi}(\text{shifted})} = \pm\pi/4$ (a direct analogue of the unit identity). For every other Gaussian prime of norm ≥ 2 , including all four associates of every split prime, $(S_\pi + S_{\bar{\pi}(\text{shifted})})/\pi$ is irrational, via an explicit closed form $S_\pi + S_{\bar{\pi}(\text{shifted})} = \arctan(b/(N+a))$ and Niven's theorem. The moduli $|c_\pi|^2$ admit elementary closed forms in the unit case $\text{Re}(w) = 0$; we conjecture no such closed form exists for $\text{Re}(w) \neq 0$ (Proposition 3.5). We propose a Class III transcendence classification via non-unitary Kummer monodromy (Theorem 6.1) and state a conjecture connecting the family to periods of CM elliptic curves.

Keywords: Gaussian primes, complex Gamma function, conservative matrix field, CM elliptic curves, Shimura conjecture, Gaussian integers, mathematical constants

1. Introduction

In two companion papers [1, 2] we identified five constants arising from anticommutative Pauli CMF matrix products and gave exact closed forms as tangents of arguments of complex Gamma function ratios. The present paper reveals the number-theoretic organising principle. The formula

$$c_\pi = |\text{Im}(z)/\text{Re}(z)|, \quad z = \frac{\Gamma(1 + w/2)}{\Gamma(1/2 + w/2)}, \quad w = \pi/N(\pi)$$

defines a well-formed constant for every Gaussian prime π . Once this formula is in hand, the remainder of the analysis is carried out entirely in the setting of complex Gamma-function ratios and the arithmetic of $\mathbb{Z}[i]$; no further reference to the CMF derivation is required. The CMF algebraic framework underlying the

construction is formalized in Lean 4 at github.com/Loublain/cmf-gaussian-primes; the analytic content of the present paper is not formalized. The structure is governed entirely by the splitting behaviour of rational primes in $\mathbb{Z}[i]$: split primes $p \equiv 1 \pmod{4}$ contribute, the ramified prime 2 contributes, and inert primes $p \equiv 3 \pmod{4}$ do not.

2. Definition and Basic Properties

A rational prime p behaves in one of three ways in $\mathbb{Z}[i]$:

- **Splits** if $p \equiv 1 \pmod{4}$: $p = \bar{\pi}\pi$ for a Gaussian prime π with $N(\pi) = p$. Examples: $5 = (2+i)(2-i)$, $13 = (3+2i)(3-2i)$.
- **Ramifies** if $p = 2$: $2 = -i(1+i)^2$, the Gaussian prime $(1+i)$ has norm 2.
- **Remains inert** if $p \equiv 3 \pmod{4}$: p itself is a Gaussian prime of norm p^2 . Examples: 3, 7, 11, 19, 23.

Definition 2.1. For a Gaussian prime $\pi \in \mathbb{Z}[i]$, define $c_\pi = |\text{Im}(z)/\text{Re}(z)|$ where $z = \Gamma(1+w/2)/\Gamma(1/2+w/2)$ and $w = \pi/N(\pi) \in \mathbb{Q}(i)$. Equivalently, $c_\pi = |\tan(S_\pi)|$ where $S_\pi = \text{Arg}(z)$ is the principal-branch argument.¹

Definition 2.2. For a Gaussian prime $\pi = a+bi$ with conjugate $\bar{\pi} = a-bi$, define the shifted angle $S_{\{\bar{\pi}(\text{shifted})\}} = \text{Arg}(\Gamma(1+\bar{w}/2)/\Gamma(3/2+\bar{w}/2))$ where $\bar{w} = \bar{\pi}/N(\pi)$. The shift refers to the denominator exponent: $1/2$ is replaced by $3/2$, equivalently $\Gamma(3/2+z) = (1/2+z)\Gamma(1/2+z)$.

2.3. Conventions and branch stability

We use the principal branch $\text{Arg} : \mathbb{C}^* \rightarrow (-\pi, \pi]$ throughout. The definition $c_\pi = |\tan(S_\pi)|$ is invariant under $S_\pi \mapsto S_\pi + \pi$ (the period of $\tan$), and admits the manifestly branch-free equivalent form $c_\pi = |\text{Im}(z)/\text{Re}(z)|$ when $\text{Re}(z) \neq 0$.

Stability lemma. Numerical evaluation at 120-digit precision (`mpmath`, `mp.dps = 120`) confirms $\text{Re}(z) > 0$ for every Gaussian prime π of norm $N(\pi) \leq 41$ (all entries of Table 1). For these primes, $\text{Arg}(z) \in (-\pi/2, \pi/2)$ strictly, so S_π lies in the interior of the principal-branch range and the additive identities of Theorems 3.3 and 3.4 involve no branch-cut crossings. We conjecture $\text{Re}(z) > 0$ for all Gaussian primes; a proof would follow from establishing that $\Gamma(1+w/2)$ and $\Gamma(1/2+w/2)$ lie in the same open half-plane through $\mathbb{C}^*$.

2.4. Relation to classical hypergeometric theory

The Beta decomposition (Theorem 3.6 below), $S_\pi = \arg(\pi) + \arg(B(w/2, 1/2))$, connects the family directly to the Euler Beta function $B(a,b) = \Gamma(a)\Gamma(b)/\Gamma(a+b)$. For the Gauss hypergeometric function, ${}_2F_1(w/2, 1/2; w/2+1/2; 1) = \Gamma(w/2 + 1/2 - 1/2)\Gamma(w/2+1/2) / [\Gamma(w/2)\Gamma(w/2+1/2-w/2+1/2)]$ is closely related to our ratio [7, §13.2]. The Möbius recursion underlying these constants (see companion paper [2]) belongs to the Poincaré–Perron family of linear difference equations with polynomial coefficients, whose general theory is treated in Jones–Thron [8]; the recursion $z_k = (r(k)z+p(k))/(p(k)z-r(k))$ is of Möbius type with polynomial r, p . The CMF framework identifies the specific family arising in this paper as the one where the Pauli anticommutation forces the involutive structure (see [2, §2.1]).

3. Structural Theorems

Theorem 3.1 (Conjugation Symmetry). $S_{\{\bar{\pi}\}} = -S_\pi$ and $|c_{\{\bar{\pi}\}}| = |c_\pi|$.

Proof. The conjugate prime $\bar{\pi}$ has $\bar{w} = \text{conj}(w)$. Since $\Gamma(\text{conj}(z)) = \text{conj}(\Gamma(z))$, $\text{Arg}(\Gamma(1+\bar{w}/2)/\Gamma(1/2+\bar{w}/2)) = -\text{Arg}(\Gamma(1+w/2)/\Gamma(1/2+w/2))$. $\square$

Theorem 3.2 (Inert Prime Theorem). For $p \equiv 3 \pmod{4}$ there is no Gaussian prime $\pi \in \mathbb{Z}[i]$ with $N(\pi) = p$, so the formula $w = \pi/N(\pi)$ does not define a constant c_π for inert primes.

Proof. $N(a+bi) = a^2+b^2=p$ would require p to be a sum of two squares. By Fermat's theorem on sums of two squares [6], p is such a sum iff $p=2$ or $p \equiv 1 \pmod{4}$. For $p \equiv 3 \pmod{4}$ no representation exists. $\square$

Theorem 3.3 (Unit Identity). $S_\pi + S_{-\pi(\text{shifted})} = \pi/4$. [Proved in [2, Theorem 4.1]: the combined Gamma ratio equals $\coth(\pi/2) \cdot (1+i)/2$, whose argument is $\pi/4$.]

Remark. Theorem 3.4 below shows that this identity extends to the associates $\pi \in \{-1+i, -1-i\}$ of the ramified prime, with the same value $\pm\pi/4$. These are the only Gaussian primes of norm ≥ 2 for which an identity of this form holds.

Theorem 3.4. Let $\pi = a + bi$ be a Gaussian prime with $N(\pi) \geq 2$ and $w = \pi/N(\pi)$. Then

$$S_\pi + S_{\bar{\pi}(\text{shifted})} = \arctan\left(\frac{b}{N(\pi) + a}\right).$$

The ratio $(S_\pi + S_{\bar{\pi}(\text{shifted})})/\pi$ is rational if and only if $\pi \in \{-1+i, -1-i\}$, in which case it equals $\pm 1/4$. For every other Gaussian prime of norm ≥ 2 — every associate of every split prime, and the associates $1+i$, $1-i$ of the ramified prime above 2 — the ratio is irrational.

Proof. Write $R(w) := \Gamma(1+w/2) \cdot \Gamma(1+\bar{w}/2) / [\Gamma(1/2+w/2) \cdot \Gamma(3/2+\bar{w}/2)]$, so that $S_\pi + S_{\bar{\pi}(\text{shifted})} = \text{Arg } R(w)$. We apply two elementary identities:

(i) $\Gamma(\bar{z}) = \overline{\Gamma(z)}$ for every z , hence $\Gamma(z) \cdot \Gamma(\bar{z}) = |\Gamma(z)|^2$ is a positive real.

(ii) $\Gamma(3/2 + \bar{w}/2) = (1/2 + \bar{w}/2) \cdot \Gamma(1/2 + \bar{w}/2)$.

Applying (i): numerator = $|\Gamma(1+w/2)|^2$, denominator = $(1/2+\bar{w}/2) \cdot |\Gamma(1/2+w/2)|^2$. Therefore

$$R(w) = \left| \frac{\Gamma(1+w/2)}{\Gamma(1/2+w/2)} \right|^2 \cdot \frac{1}{1/2 + \bar{w}/2}.$$

The first factor is a positive real, so $\text{Arg } R(w) = -\text{Arg}(1/2+\bar{w}/2)$. For $w = (a+bi)/N$, we have $1/2+\bar{w}/2 = (N+a-bi)/(2N)$. Since $N+a > 0$ (as $N \geq 2$ forces $N > |a|$), its argument is $-\arctan(b/(N+a))$, giving the closed form.

For the rationality classification, Niven's theorem (1956) states that $\arctan(q)/\pi$ is rational for $q \in \mathbb{Q}$ only when $q \in \{0, \pm 1\}$. The case $b = 0$ is impossible for norm ≥ 2 non-inert primes. The cases $q = \pm 1$ give $b = \pm(N+a)$, i.e. $(a + 1/2)^2 + (b \mp 1/2)^2 = 1/2$, a circle of radius $1/\sqrt{2}$. The integer points on this circle are $(0,0)$, $(0,\pm 1)$, $(-1,0)$, $(-1,\pm 1)$. Excluding non-primes and units leaves exactly $\pi \in \{-1+i, -1-i\}$. $\square$

Proposition 3.5 (Hyperbolic Closed Form, Unit Case). Let $\pi \in \mathbb{Z}[i]$ be a Gaussian prime or unit, $w = \pi/N(\pi)$, and $R(w) = \Gamma(1+w/2)/\Gamma(1/2+w/2)$.

(a) If $\text{Re}(w) = 0$ — equivalently $\pi = \pm i$ — then

$$|R(w)|^2 = \frac{w}{2i} \cdot \coth\left(\frac{\pi w}{2i}\right),$$

which equals $(1/2) \coth(\pi/2)$ for $\pi = \pm i$.

- (b) If $\text{Re}(w) \neq 0$, no expression for $|R(w)|^2$ has been found inside the ring generated over $\mathbb{Q}(\pi)$ by $\{\cosh(q\pi), \sinh(q\pi) : q \in \mathbb{Q}\}$. Extensive PSLQ searches to 120 decimal digits with coefficient bound 10^6 are null for every Gaussian prime of norm ≤ 41 (§6.5). We conjecture that no such expression exists.

Proof of (a). For $w = it$, $t \in \mathbb{R}$, the reflection formula $\Gamma(z)\Gamma(1-z) = \pi/\sin(\pi z)$ at $z = 1/2 + it/2$ gives $|\Gamma(1/2+it/2)|^2 = \pi/\cosh(\pi t/2)$; combined with $\Gamma(z+1) = z\Gamma(z)$ at $z = it/2$ it gives $|\Gamma(1+it/2)|^2 = \pi t/(2 \sinh(\pi t/2))$. Dividing yields $(t/2) \coth(\pi t/2)$, matching the stated formula with $w = it$. $\square$

Remark (structural obstruction underlying (b)). The toolkit for closed-form evaluation of $|\Gamma(z)|^2$ in $\mathbb{R}$ consists of three classical Γ -identities — reflection $\Gamma(z)\Gamma(1-z) = \pi/\sin(\pi z)$, integer translation $\Gamma(z+1) = z\Gamma(z)$, and Gauss multiplication

$$\prod_{k=0}^{n-1} \Gamma(z + k/n) = (2\pi)^{(n-1)/2} n^{1/2-nz} \Gamma(nz)$$

— applied to the identity $|\Gamma(z)|^2 = \Gamma(z)\Gamma(\bar{z})$.

Reflection and translation alone connect $\Gamma(z)$ to $\Gamma(\bar{z})$ precisely when $\bar{z}$ lies in the orbit of z under $z \mapsto z+1$ and $z \mapsto 1-z$, i.e. when $\text{Re}(z) \in \frac{1}{2}\mathbb{Z}$. For $\text{Re}(w) \neq 0$, both $z = 1 + w/2$ and $z = 1/2 + w/2$ fail this condition, so $|\Gamma(z)|^2$ for these arguments is not reducible to elementary functions of πw by reflection and translation alone.

Gauss multiplication at order n introduces $n-1$ additional Γ -values $\Gamma(z+k/n)$, each with $\text{Im}(z)$ unchanged. The reflection orbit then extends to include $(1 - \text{Re}(z) - k/n + \mathbb{Z})$ for $k = 0, \dots, n-1$, so $\bar{z}$ enters the orbit when $\text{Re}(z) \in \frac{1}{2n}\mathbb{Z}$, attainable at $n = N$ for $w = (a + bi)/N$. However, the Gauss identity is a single multiplicative relation among $n+1$ Γ -values (including $\Gamma(nz)$), not an isolation of any individual $|\Gamma(z+k/n)|^2$. Inverting it to express $|\Gamma(z)|^2$ alone reintroduces $n-1$ other Γ -values at non-half-integer real parts, all subject to the same obstruction.

An extended PSLQ search incorporating $\cosh(q\pi)$ and $\sinh(q\pi)$ at $q = k/(2N)$ (the denominators arising from Gauss multiplication at order N) returns null at 100 decimal digits for $\pi \in \{2+i, 3+2i, 4+i\}$, consistent with this obstruction.

A rigorous non-existence proof — ruling out closed forms in $\mathbb{R}$ via *any* algebraic identity, not merely the classical toolkit — lies in the domain of transcendence theory and is outside the scope of this paper. We conjecture no closed form exists in $\mathbb{R}$ but do not prove this.

Remark 3.5.1. Proposition 3.5 concerns the individual Gamma ratio $\Gamma(1+w/2)/\Gamma(1/2+w/2)$. The bilinear combination with its complex conjugate that appears in Theorem 3.4 is elementary for every Gaussian prime π with $N(\pi) \geq 2$ (it equals $1/(1/2+\bar{w}/2)$, up to the real positive factor $|\Gamma(1+w/2)/\Gamma(1/2+w/2)|^2$). This illustrates that the proposed Class III nature of the individual constants S_π is compatible with elementary closed forms for specific symmetric combinations. The proposed Class III classification in §6 applies to the individual S_π (and to their Hurwitz-zeta derivatives, by §6.5), not to every algebraic combination thereof.

Theorem 3.6 (Beta Decomposition). $S_\pi = \arg(\pi) + \arg(B(w/2, 1/2))$ where $B(a,b) = \Gamma(a)\Gamma(b)/\Gamma(a+b)$.

Proof. $\Gamma(1+w/2) = (w/2)\Gamma(w/2)$, so $\Gamma(1+w/2)/\Gamma(1/2+w/2) = (w/2) \cdot B(w/2, 1/2)/\sqrt{\pi}$. Taking arguments: $S_{-\pi} = \arg(w/2) + \arg(B) = \arg(\pi) + \arg(B)$. $\square$

4. Computed Values

Table 1 gives the first members of the family, computed to 18 significant figures using mpmath (mp.dps = 80). Each value has been verified absent from the OEIS, Wolfram Alpha, and the ISC database at the precision tested.

π	$N(\pi)$	Rational prime behaviour	$c_{-\pi}$
i	1	unit	0.554996111573619503...
$-i$	1	unit	0.554996111573619503...
$1+i$	2	2 ramified	0.207879576350761908...
$-(1+i)$	2	2 ramified	0.655794202632672435...
$2+i$	5	5 split, 1 mod 4	0.092546110069323602...
$-(2+i)$	5	5 split, 1 mod 4	0.247582218708610699...
$3+2i$	13	13 split, 1 mod 4	0.083005998252748272...
$3-2i$	13	13 split, 1 mod 4	0.083005998252748272...
$4+i$	17	17 split, 1 mod 4	0.031720900524284986...
$5+2i$	29	29 split, 1 mod 4	0.039548684965468606...
$6+i$	37	37 split, 1 mod 4	0.015673823447039764...
$5+4i$	41	41 split, 1 mod 4	0.058872673342944218...

Table 1. Gaussian prime constants to 18 significant figures.

5. Reconciliation with the Original Five Constants

The five constants of [1, 2] appear in Table 1: $c_1 = c_{-i}$, $c_4 = c_{-(2+i)}$, $c_5 = c_{1+i}$. The constant c_3 uses a mixed-shift formula related to c_{-i} (Theorem 3.3). The constant c_2 is composite: its closed form involves $\sqrt{1 - 8i}$, whose norm is $65 = 5 \times 13$, connecting it to the Gaussian primes $(2+i)$ and $(3+2i)$ simultaneously. Specifically $S_2 = S(r_1) + S(r_2)$ where r_1, r_2 are roots of $4iz^2 + (1-4i)z + i = 0$, and these roots live in the ring class field compositum $\mathbb{Q}(i, \sqrt{1+2i}, \sqrt{3+2i})$.

6. Position in the Transcendence Hierarchy

Status of this section. Section 6 separates into two registers. Theorem 6.1 is a proved result: for every Gaussian prime π with $N(\pi) \geq 1$, the constant $S_{-\pi}$ is not an exponential period in the Fresán–Jossen sense, because the associated Kummer D-module has monodromy of infinite order. The three-class hierarchy of §6.1, the conjectural Galois-theoretic heuristic of §6.3, and Conjecture 6.2 are offered as interpretive framework: they describe where our constants appear to sit relative to existing motivic theory, and what properties an extended theory would need. The proposed category of algebraic non-rational Kummer twists does not yet exist in the literature; the Galois-group suggestion in §6.3 is heuristic; the PSLQ evidence of §6.5 is consistent with the framework but does not establish it.

Theorem 6.1 (Classification Obstruction). $S_{-\pi}$ is not an exponential period in the sense of Fresán–Jossen

[9] for any Gaussian prime π with $N(\pi) \geq 1$.

Proof. The integral $\Gamma(w/2) = \int t^{w/2-1} e^{-t} dt$ corresponds to D-module $E(1) \otimes K(a)$ on $\mathbb{G}_m$ with $a = w/2-1$ and Kummer monodromy $\mu = e^{(\pi i w)}$. Since $w \in \mathbb{Q}(i) \setminus \mathbb{Q}$, $\text{Im}(w) \neq 0$ and $|\mu| = e^{(-\pi \cdot \text{Im}(w))} \neq 1$. Non-unitary monodromy has infinite order. An algebraic cover of degree d pulls back $K(a)$ to $K(da)$, trivial only if $a \in \mathbb{Q}$. But $a \in \mathbb{Q}(i) \setminus \mathbb{Q}$, so no cover exists. $\square$

6.1. The three-class hierarchy

An exponential period in the sense of Fresán–Jossen [9] is an integral of the form $\int_{\gamma} e^{-f(x)} \omega(x)$, where f is a regular function and ω is an algebraic differential form on a variety over $\bar{\mathbb{Q}}$, and γ is a semialgebraic cycle. Classical Gamma values $\Gamma(a+1)$ with $a \in \mathbb{Q}$ are exponential periods because the underlying rank-1 D-module $E(1) \otimes K(a)$ on $\mathbb{G}_m$ has Kummer monodromy $e^{(2\pi i a)}$ of finite order, trivialised by the algebraic cover $z \mapsto z^d$ for any d with $da \in \mathbb{Z}$. Our constants involve Γ at $a = w/2-1 \in \mathbb{Q}(i) \setminus \mathbb{Q}$, for which $|e^{(2\pi i a)}| = e^{(-\pi \text{Im}(w))} \neq 1$ (Theorem 6.1 above): the monodromy has infinite order, no algebraic cover trivialises $K(a)$, and the associated D-module lies outside the exponential motive category.

Class I ($a \in \mathbb{Z}$): $\Gamma(a+1) = a!$ rational.

Class II ($a \in \mathbb{Q} \setminus \mathbb{Z}$): monodromy a root of unity; $\Gamma(a+1)$ an exponential period. Includes $\Gamma(1/4)$, $\Gamma(1/3)$, $\sqrt{\pi}$, all Chowla–Selberg values.

Class III (proposed) ($a \in \bar{\mathbb{Q}} \setminus \mathbb{Q}$): $|\mu| \neq 1$; not an exponential period. Our constants have $a = w/2-1 \in \mathbb{Q}(i) \setminus \mathbb{Q}$, placing them in the proposed Class III; no existing motivic category directly accommodates them.

6.2. Monodromy data

π	N	w	$a = w/2-1$	$ \mu $
i	1	i	$-1+i/2$	$e^{-\pi} \approx 0.0432$
$1+i$	2	$(1+i)/2$	$-3/4+i/4$	$e^{-\pi/2} \approx 0.208$
$2+i$	5	$(2+i)/5$	$-4/5+i/10$	$e^{-\pi/5} \approx 0.534$
$3+2i$	13	$(3+2i)/13$	$-11/13+i/13$	$e^{-2\pi/13} \approx 0.617$
$4+i$	17	$(4+i)/17$	$-13/17+i/34$	$e^{-\pi/17} \approx 0.831$

Table 2. Kummer monodromy. As $N(\pi) \rightarrow \infty$, $|\mu| \rightarrow 1$ but never equals 1.

6.3. Conjectural Galois structure

For Class II gamma motives, the Galois group contains μ_n (finite). For the proposed Class III, μ_n would be replaced by $\mathbb{G}_m$ (infinite-order monodromy), suggesting one additional algebraic independence should hold — consistent with all PSLQ failures.

6.4. Why classical tools fail

Chowla–Selberg handles Γ at rational real arguments (Class II). Shimura/Deligne handle Class II motives. Hecke L-functions with trivial conductor yield no PSLQ match. All probe Class I–II; our constants are

proposed to sit in Class III.

6.5. Extended PSLQ evidence from the Hurwitz-zeta family

We extended the PSLQ search to the family

$$S_\pi(n) = \text{Im}[\zeta'(-n, 1 + w/2) - \zeta'(-n, 1/2 + w/2)], \quad n = 0, 1, 2, 3, 4$$

for $\pi \in \{i, 1+i, 2+i, -(2+i), 3+2i, 4+i, 5+2i\}$ at 120-digit precision (mpmath mp.dps = 120, Bailey–Broadhurst pslq, maxcoeff = 10^6). For $n=0$ this reduces to S_π via the Lerch formula (verified). Three search configurations were run:

- *Within- π* : $\{S_\pi(0), \dots, S_\pi(4)\}$ + standard basis, for each of the 7 primes.
- *Across- π at fixed n* : all 7 values $\{S_\pi(n)\}$ + standard basis, for $n = 0, \dots, 4$.
- *Mixed*: all 35 values together with the standard basis (41 elements).

All 20 searches returned null. Extended bases incorporating $\cosh(q\pi)$, $\sinh(q\pi)$ at $q = k/(2N)$ — the denominators arising from Gauss multiplication at order N — were also tested for $\pi \in \{2+i, 3+2i, 4+i\}$ at 100 decimal digits and returned null (see Remark following Proposition 3.5). This is the expected outcome under the proposed Class III classification. Full details at github.com/Loublain/cmf-gaussian-primes [archived: doi:10.5281/zenodo.19670531].

Conjecture 6.2. The constants S_π are natural candidates for periods of an extended motivic category incorporating D-modules $E(1) \otimes K(a)$ with $a \in \mathbb{Q}(i) \setminus \mathbb{Q}$ — ‘algebraic non-rational Kummer twists’. Such a category does not yet exist.

7. Computational Notes

All constants computed with mpmath at 80–160 decimal digits. The branch-free evaluation form $c_\pi = |\text{Im}(z)/\text{Re}(z)|$ is used throughout.

```
import mpmath; mpmath.mp.dps = 80
def c_pi(pi_gauss):
    w = pi_gauss / abs(pi_gauss)**2
    z = mpmath.gamma(1+w/2) / mpmath.gamma(0.5+w/2)
    return abs(mpmath.im(z) / mpmath.re(z)) # branch-free
print(c_pi(1+1j)) # 0.20788...
print(c_pi(3+2j)) # 0.08301...
```

8. Open Questions

- (1) Prove Conjecture 6.2.
- (2) Find a product formula for the composite constant c_2 .
- (3) Extend to Eisenstein integers $\mathbb{Z}[\omega]$ and other imaginary quadratic fields.

- (4) Characterise constants at $w = 1/p$ for inert primes.
- (5) Connect the norm-5 constants $c_{\{\pm(2+i)\}}$ to the $\zeta(5)$ obstruction.

9. Conclusion

We have identified an infinite family of mathematical constants $c_\pi = |\text{Im}(z)/\text{Re}(z)|$ with $z = \Gamma(1+w/2)/\Gamma(1/2+w/2)$ and $w = \pi/N(\pi)$, one per Gaussian prime of $\mathbb{Z}[i]$, each verified absent from standard databases. Split primes contribute, inert primes do not. The unit identity $S_1 + S_3 = \pi/4$ extends by Theorem 3.4 to two associates of the ramified prime $(-1+i$ and $-1-i)$, where the same combination gives $\pm\pi/4$. All other Gaussian primes of norm ≥ 2 produce irrational ratios $(S_\pi + S_{\{\bar{\pi}(\text{shifted})\}})/\pi$ via the closed form $\arctan(b/(N+a))$ and Niven’s theorem. This is an elementary result; the proposed Class III classification (§6) applies to the individual S_π , not to this specific bilinear combination. Theorem 6.1 is a proved obstruction from the Fresán–Jossen framework; the three-class hierarchy of §6.1 and Conjecture 6.2 are proposed as interpretive framework pending the development of an extended motivic category.

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Acknowledgements. The author thanks Claude Sonnet 4.6 (Anthropic) for assistance with derivation, computation, and manuscript preparation throughout this research. Correspondence: louisblaine45@gmail.com

Footnotes

¹ The base CMF structure from which these constants are derived is formalized in `CMF.lean`. The conservativity condition is machine-verified. See github.com/Loublain/cmf-gaussian-primes